

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0517

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Brightness Image

A **low brightness image** is an image that appears very dark because it has been captured under poor lighting conditions or with improper camera settings. In such images, most of the pixel intensity values lie in the lower end of the grayscale range (close to 0). As a result, objects, edges, and textures become difficult to see. Low brightness is one of the most common image quality problems in digital image processing because it affects both human viewing and computer vision applications.

For example:

- A photograph taken at night without flash.
- A CCTV camera recording in a dimly lit room.
- A medical X-ray or MRI image with insufficient illumination.
- A satellite image captured under cloudy weather.
- A mobile phone image taken indoors with poor lighting.

Low brightness images need enhancement before further processing such as segmentation, feature extraction, edge detection, or object recognition.

(a) Interpret the scenario and explain the cause of low brightness

Scenario Interpretation

Suppose a security camera captures an image at night. Since very little light reaches the camera sensor, the resulting image appears dark and important objects such as people, vehicles, or doors are difficult to recognize. This is called a **low brightness image**.

The image has low visibility because the intensity values of most pixels are concentrated in the dark region instead of being distributed across the entire grayscale range.

What is Brightness?

Brightness refers to the amount of light present in an image.

- **High brightness** → Image appears bright.
- **Low brightness** → Image appears dark.

In grayscale images, brightness values range from:

- **0 = Black**
- **255 = White**

A low brightness image contains more pixel values close to **0** than **255**.

Causes of Low Brightness

1. Insufficient Lighting

This is the most common reason.

Examples:

- Night photography
- Dark rooms
- Underground parking
- Poor weather conditions

Very little light reaches the camera sensor.

2. Incorrect Camera Exposure

Exposure determines how much light reaches the sensor.

If exposure is too low:

- Image becomes dark
- Details disappear

3. Fast Shutter Speed

A fast shutter closes quickly.

Advantages:

- Reduces motion blur

Disadvantage:

- Allows less light into the camera.

Hence, the image becomes darker.

4. Small Aperture

Aperture controls the opening of the camera lens.

Small aperture:

- Less light enters
- Lower brightness

Large aperture:

- More light enters
- Brighter image

5. Low ISO Value

ISO controls sensor sensitivity.

Low ISO:

- Less sensitive to light
- Dark image

High ISO:

- Brighter image
- May introduce noise

6. Shadow Regions

Objects hidden inside shadows receive very little light.

Therefore,

- Details disappear
- Contrast decreases

7. Sensor Limitations

Cheap or old camera sensors cannot capture enough light in dark environments.

Hence,

- Image appears dim
- Noise increases

Characteristics of Low Brightness Images

- Very dark appearance
- Low contrast
- Hidden details
- Poor visibility
- Difficult object recognition
- Large number of black pixels
- Poor image quality

(b) Identify key issues affecting image visibility

A low brightness image suffers from several problems.

1. Poor Contrast

Contrast is the difference between bright and dark regions.

Low brightness reduces this difference.

Example:

Dark object on dark background becomes invisible.

Result:

- Difficult to distinguish objects.

2. Loss of Detail

Small image details disappear.

Examples:

- Human face
- Vehicle number plate
- Tree leaves
- Building edges

Fine textures cannot be identified.

3. Hidden Objects

Objects become invisible because their intensity values are very similar to the background.

Example:

A black cat in a dark room.

Both have similar brightness.

4. Increased Noise

When brightness is increased later, sensor noise also becomes visible.

Noise appears as:

- Random white dots
- Random black dots

Noise decreases image quality.

5. Weak Edges

Edge detection algorithms depend on intensity differences.

Since brightness is very low,

Edges become weak.

Algorithms like:

- Sobel
- Prewitt
- Canny

produce poor results.

6. Difficulty in Segmentation

Image segmentation separates objects from the background.

Low brightness causes:

- Incorrect boundaries
- Missing objects
- Poor segmentation accuracy

7. Poor Object Recognition

AI systems may fail to recognize objects because important features are hidden.

Examples:

- Face recognition
- Fingerprint recognition
- Vehicle detection
- Medical diagnosis

8. Reduced Visual Quality

Humans also struggle to understand the image.

Objects appear blurred.

Image becomes difficult to interpret.

Summary of Issues

Issue	Effect
Poor Contrast	Difficult to separate objects
Loss of Details	Small features disappear
Hidden Objects	Objects merge with background
Noise	Grainy appearance
Weak Edges	Poor edge detection
Poor Segmentation	Incorrect object boundaries
Recognition Failure	AI accuracy decreases
Low Visual Quality	Difficult for human viewing

(c) Apply an appropriate enhancement technique

Several enhancement methods can improve low brightness images.

Examples:

- Brightness Adjustment
- Contrast Stretching
- Gamma Correction
- Adaptive Histogram Equalization (CLAHE)
- Histogram Equalization

Among these,

Histogram Equalization is one of the best techniques.

Histogram Equalization

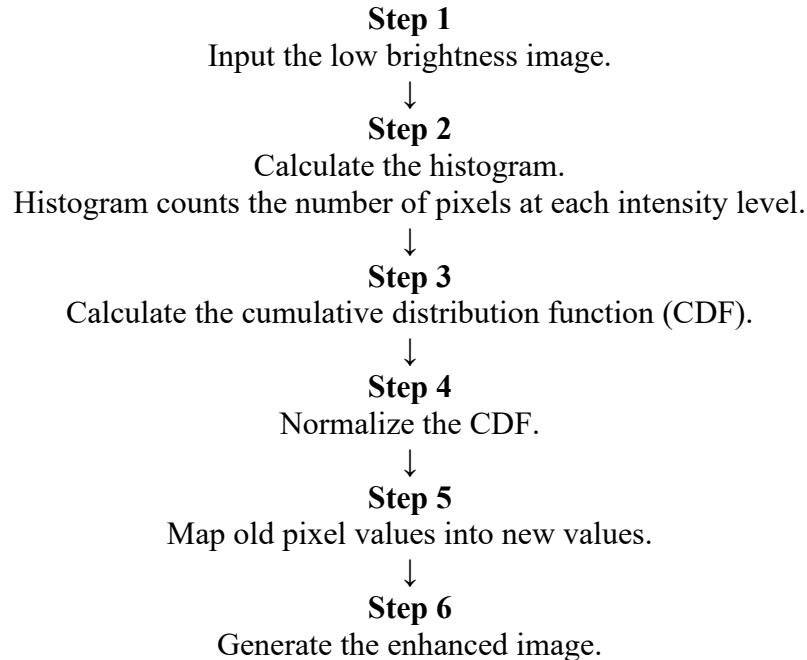
Histogram Equalization improves contrast by redistributing pixel intensity values over the complete grayscale range.

Instead of keeping pixels concentrated in dark regions,
it spreads them from

0 → 255

making the image brighter.

Working Steps



Example

Original Image

40 45 50

55 60 65

70 75 80

Most values are dark.

Histogram

Dark Pixels 

Bright Pixels 

Most pixels lie near zero.

After Histogram Equalization

20 55 90

125 160 190

215 235 255

Now pixel values cover the entire intensity range.

Enhanced Histogram

Dark 

Medium 

Bright 

Pixel distribution becomes more uniform.

Advantages

- Improves brightness
- Improves contrast
- Reveals hidden details
- Better edge detection
- Better segmentation
- Better object recognition
- Easy to implement
- Fast processing

Disadvantages

- May amplify image noise.
- Can make some areas appear over-bright.
- Not suitable for all images, especially when local contrast enhancement is needed.

(d) Justify your choice with reasoning

Histogram Equalization is selected because it is one of the simplest and most effective techniques for enhancing low brightness images.

Reasons

1. Improves Overall Brightness

Dark images become brighter.

Objects become easier to observe.

2. Enhances Contrast

Pixel intensities spread across the full grayscale range.

Objects become distinguishable from the background.

3. Reveals Hidden Details

Textures

Edges

Patterns

Previously invisible information becomes visible.

4. Improves Image Analysis

Algorithms perform better.

Examples:

- Edge Detection
- Thresholding
- Segmentation
- Face Recognition
- OCR
- Medical Diagnosis

5. Automatic Technique

No manual brightness adjustment is required.
Works automatically using histogram statistics.

6. Computationally Efficient

Requires little processing time.

Suitable for:

- Mobile devices
- CCTV systems
- Medical imaging
- Satellite imaging

7. Widely Used

Used in:

- Medical image enhancement
- Remote sensing
- Traffic monitoring
- Surveillance
- Industrial inspection
- Machine vision

(e) Present your explanation clearly

A low brightness image is produced when insufficient light reaches the camera sensor because of poor lighting conditions or improper camera settings. Such images appear dark, have poor contrast, and lose important details, making it difficult for humans and computer vision systems to interpret them. The main issues include poor visibility, weak edges, hidden objects, increased noise, and reduced recognition accuracy. To overcome these problems, **Histogram Equalization** is applied. This technique redistributes pixel intensity values across the full grayscale range, thereby increasing brightness and improving contrast. As a result, hidden details become visible, image quality is enhanced, and subsequent image processing tasks such as edge detection, segmentation, and object recognition become more accurate.

Real-World Applications

- Night vision systems
- CCTV surveillance
- Medical imaging (X-ray, MRI, CT scan)
- Satellite and aerial image enhancement
- Autonomous vehicles
- Robotics and computer vision
- Mobile phone cameras
- Industrial quality inspection

Conclusion

Low brightness images are caused by poor illumination, incorrect camera settings, shadows, or sensor limitations. These images suffer from poor contrast, hidden details, weak edges, increased noise, and reduced visual quality. **Histogram Equalization** is a suitable enhancement technique because it redistributes pixel intensities over the complete grayscale range, improving brightness and contrast while revealing hidden information. This enhancement significantly improves both human visual perception and the performance of image processing and computer vision applications.

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand